

■ PROCESSORS

AMD Introduces Ryzen AI 300 and 9000 Series Processors at Computex 2024

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At Computex 2024, AMD announced the launch of its next-generation Ryzen AI 300 and 9000 Series desktop processors. These processors are designed to enhance performance, efficiency, and AI capabilities for both laptops and desktop PCs. AMD had recently launched their Ryzen 7040 series processors for laptops with an inbuilt NPU for accelerating AI workloads and today's launch falls in line with the push for 'AI' PCs in the market. Qualcomm's Snapdragon X Elite currently leads the roost with 45 TOPS NPU performance and that's being challenged by the likes of AMD and Intel. The AMD Ryzen AI 300 processors claim to achieve 50 TOPS which is a huge jump in performance from its predecessor.

AMD Ryzen AI 300 Series notebook processors

The AMD Ryzen AI 300 Series processors, based on the new AMD XDNA 2 architecture, aim to make significant advancements in the AI PC market. These processors feature a Neural Processing Unit (NPU) with 50 TOPS (trillions of operations per second) of AI processing power, positioning AMD's latest offering above the requirements for AI-enhanced PCs, such as the Copilot+ AI PCs.

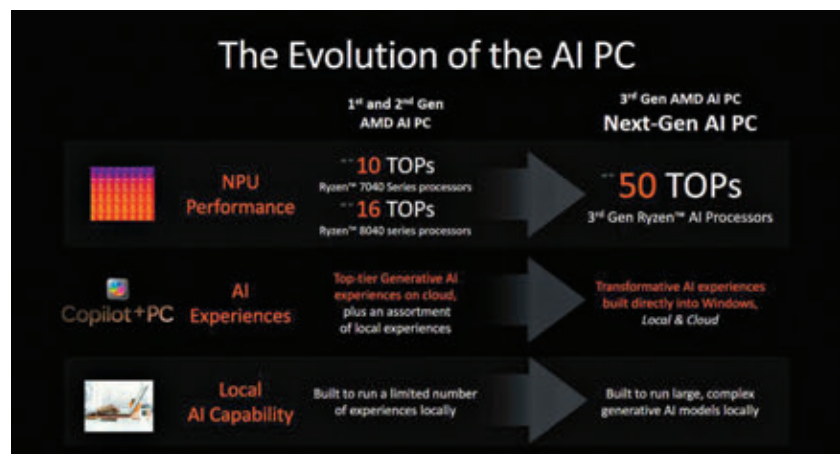
These new AMD Ryzen AI 300 series processors feature up to 12 Cores and 24 Threads and there's about 50 per cent more on-die L3 cache memory. Each of the new Ryzen AI 300 processors also sport an inbuilt GPU based on their own RDNA 3.5 graphics architec-

ture. These new Radeon 800M IGPs sport up to 16 Compute Units which is comparable to some of the entry-level desktop graphics cards from a couple of generations ago.

Features

This is not the first time that AMD has introduced an NPU in their processors. As mentioned previously, the AMD

which provide leadership AI and compute performance for ultrathin and premium Copilot+ PCs," said Jack Huynh, senior vice president and general manager, Computing and Graphics Group at AMD. "The AMD Ryzen AI 300 series processors push the limits on what's possible featuring the fastest APU performance6, the world's most powerful NPU with 50 TOPs, and the world's



Ryzen 7040 were the first notebook chips with inbuilt NPUs capable of hitting 10 TOPS. This was followed by the Ryzen 8040 series that took it up a notch to about 16 TOPS. And today's announcement will see the new Ryzen AI 300 series processors hit 50 TOPS. This would make it the most powerful notebook AI processor.

AMD Ryzen AI 300 Processor AI Performance compared

"We are so excited to introduce the Ryzen 9000 series, the world's most powerful desktop processors for gamers and creators, and the 3rd generation AMD Ryzen AI processors,

first block floating point NPU doubling the performance of 16-bit applications without sacrificing accuracy."

"We are in the midst of a massive AI platform shift, with the promise to transform how we live and work. That's why our deep partnership with AMD, which has spanned multiple computing platforms, from the PC to custom silicon for Xbox, and now to AI, is so important to us," said Satya Nadella, Chairman and CEO of Microsoft. 

